Assignment: Pipeline Hazards

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Grading (total 4 points)

- Question 1: 0.6 points
- Question 2: 0.4 points (each part 0.2 point)
- Question 3: 0.4 points
- Question 4: 2 points (each part 0.4 point)
- Question 5: 0.4 points
- Question 6: 0.2 points

Question 1: Dependencies

Find the data dependencies between all the instructions below. Write your answers in the format of the given example.

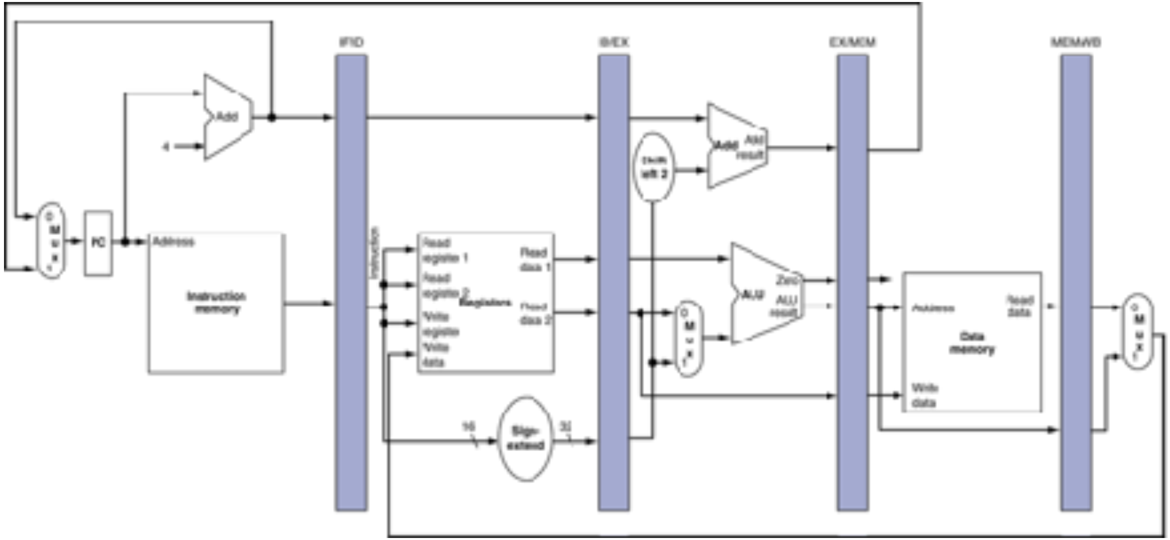
- For example: I2 has a data dependency with I1 in R2.

```
I1. sub   R2, R5, R4
I2. add   R4, R2, R5
I3. lw    R5, 100(R2)
I4. sub   R2, R5, R4
I5. sw    R2, 101(R2)
```

Question 2: EX forwarding paths

A. Add the necessary forwarding paths for the following instructions in the datapath shown here. .

```
A
1 add   R3, R3, R4
2 sub   R4, R2, R3
3 add   R4, R5, R5
```

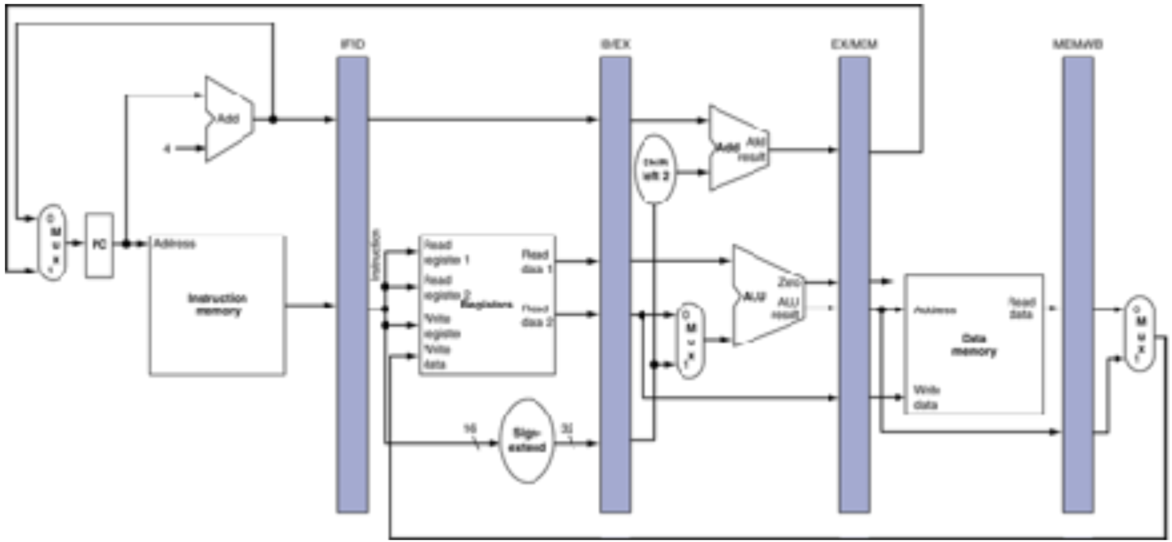


Question 2: EX forwarding paths

B. Add the necessary forwarding paths for the following instructions in the datapath shown here.

B

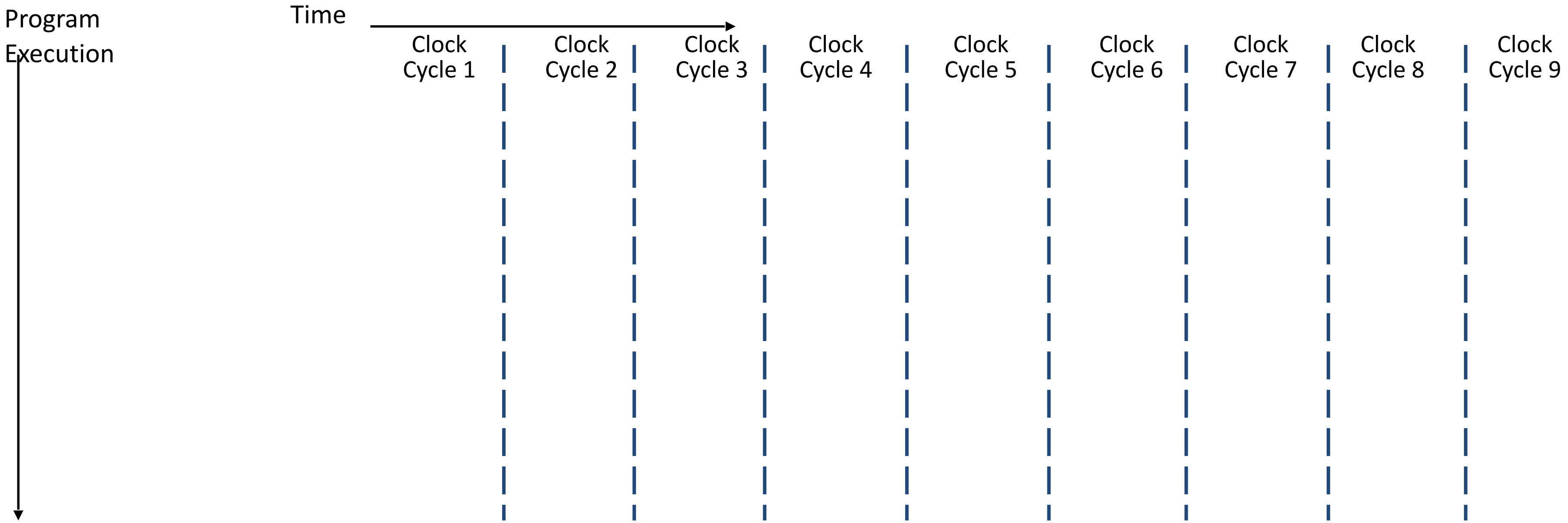
```
1  add   R3, R3, R4
2  sub   R4, R2, R4
3  add   R4, R5, R3
```



Question 3: Performance with stalls

How long will the following code take on a 5-stage MIPS pipeline with forwarding and appropriate NOPs? A. Fill out the pipeline and draw the forwarding path in the pipeline. B. How many cycles it takes to execute all three instructions?

```
add  R5, R5, R7
lw   R6, 100(R7)
sub  R7, R6, R8
```



Question 4: Load-to-use delay

The code below was written without regard for the delay slots.

(There may be NOPs missing. The pipeline has a double-pumped register file and forwarding.)

```
lw   R15, 0(R2)
add  R14, R15, R15
lw   R16, 4(R2)
add  R17, R16, R16
```

Answer Part A, B, C, D, and E.

Question 4: Load-to-use delay

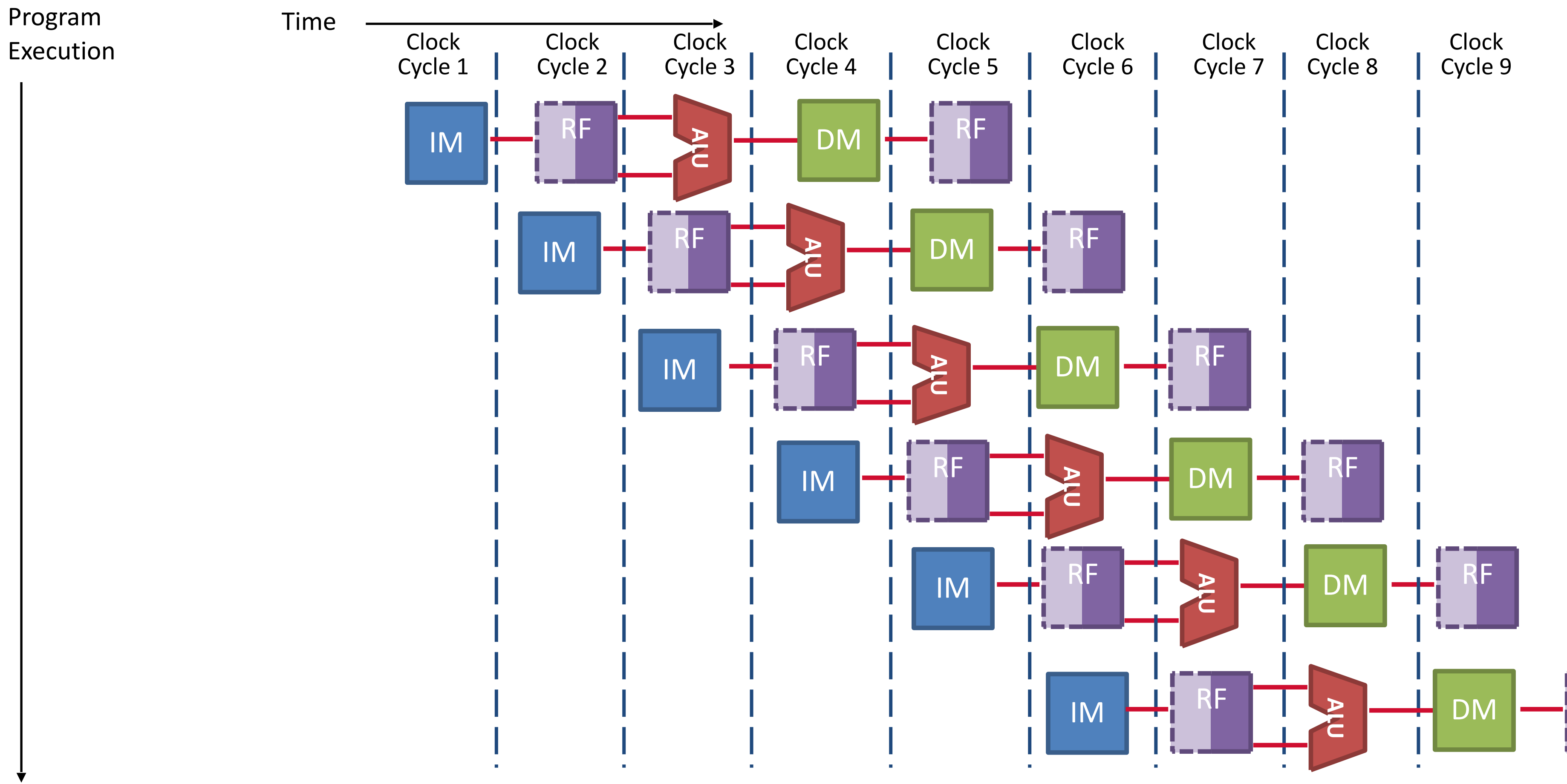
A) Identify dependencies

```
lw    R15, 0(R2)
add   R14, R15, R15
lw    R16, 4(R2)
add   R17, R16, R16
```

B) Insert NOPs to resolve the dependencies

C) After adding NOPs in Part B, add the instructions and draw the forwarding path(s) to the pipeline.

Question 4: Load-to-use delay

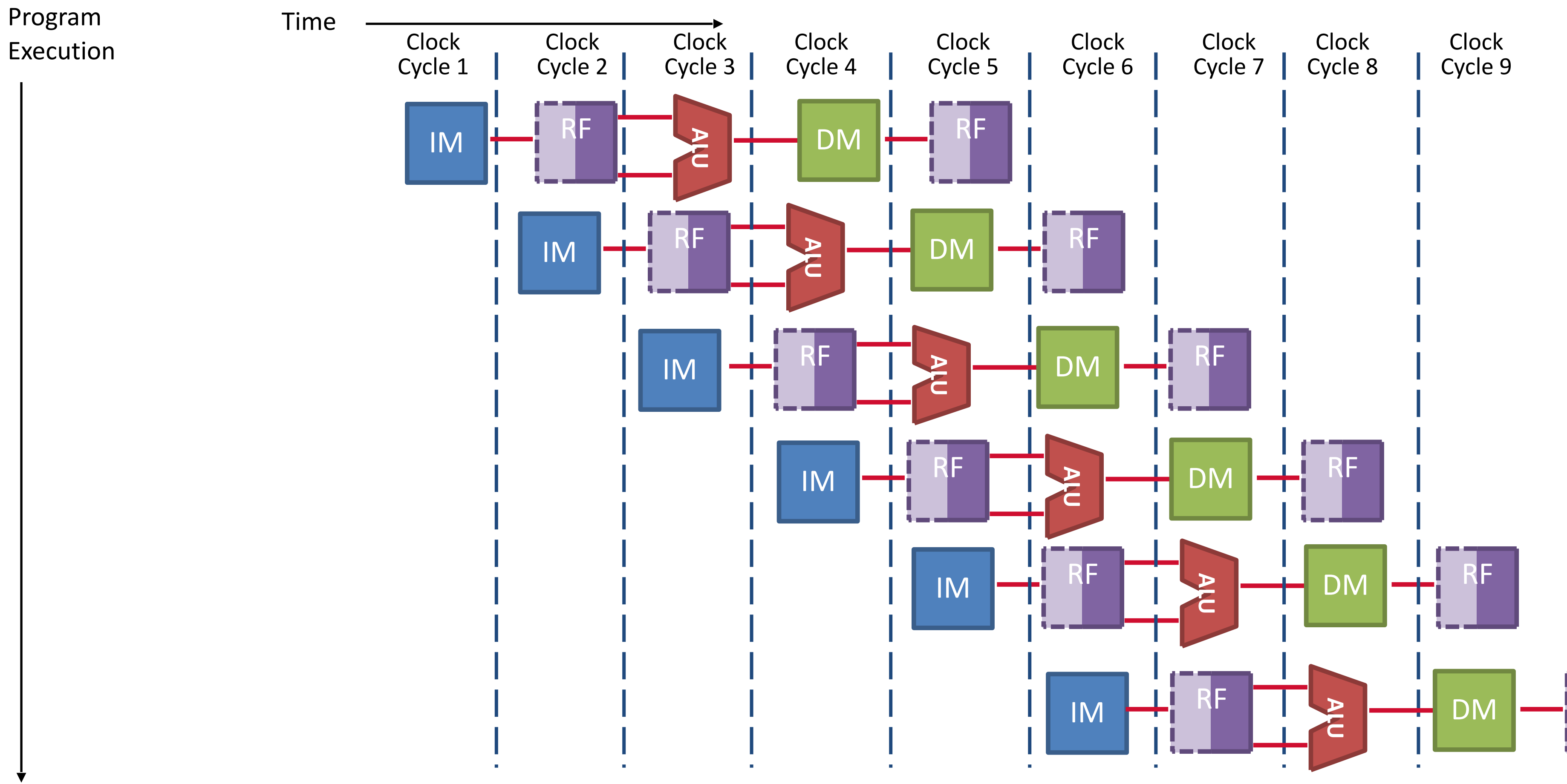


Question 4: Load-to-use delay

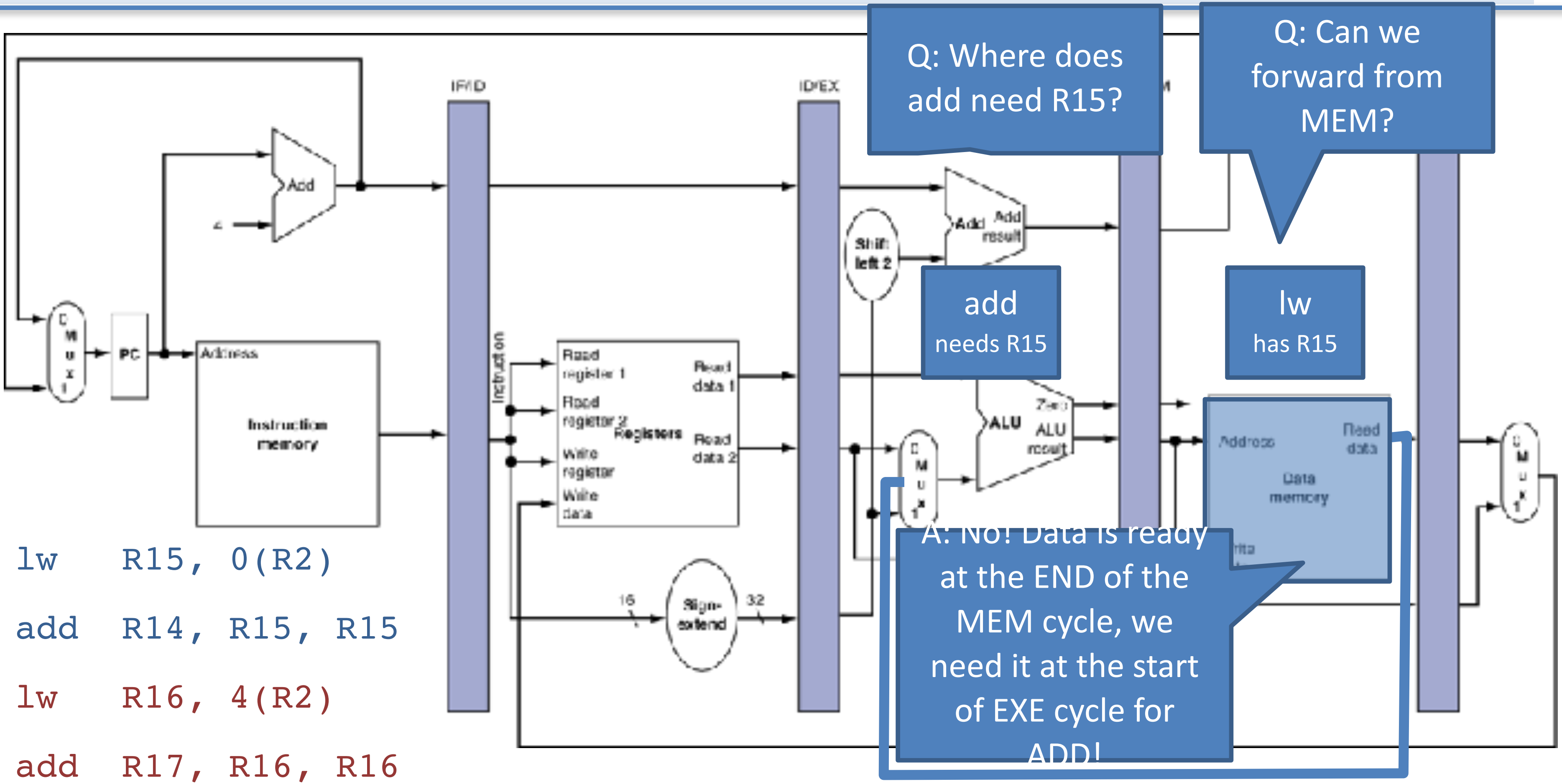
D) We would like to keep the pipeline full as much as possible and improve the performance. Can we reorder the instructions to avoid NOPs? If so, how?

E) If the answer to Part D is yes, then add the reordered instructions and draw the forwarding path(s) in the pipeline.

Question 4: Load-to-use delay



Hint: Load-to-use delay




```
lw    R15, 0(R2)
add   R14, R15, R15
lw    R16, 4(R2)
add   R17, R16, R16
```

Question 5: MEM-to-MEM stalls

Fill in the pipeline for the following code assuming forwarding and double-pumped registers. Show the forwarding path(s) in the pipeline. If there is a dependency that cannot be resolved with forwarding, then add **bubbles** to the pipeline.

```
add   R4, R5, R2
lw    R15, 0(R4)
sw    R15, 4(R2)
```

Question 5: MEM-to-MEM stalls

Program Execution

Time

Clock Cycle 1	Clock Cycle 2	Clock Cycle 3	Clock Cycle 4	Clock Cycle 5	Clock Cycle 6	Clock Cycle 7	Clock Cycle 8	Clock Cycle 9

Question 6: Branch delay slot

The code in Part A has two data dependencies (R3 and R4), and can be resolved using forwarding and double-pumped register file. However, we still have control dependency for “beq” instruction. To solve the control dependency issue, we add a “nop” after the “beq” instruction, as shown in Part B.

Question: We would like to keep the pipeline full as much as possible and improve the performance. Can we reorder the instructions to avoid NOPs? If so, how?

A) Identify dependencies

```
add  R3, R3, R4
sub  R4, R2, R3
beq  R0, R2, end
addi R4, R4, 12
end:
```

B) insert NOPs

```
add  R3, R3, R4
sub  R4, R2, R3
beq  R0, R2, end
nop
addi R4, R4, 12
```

C) reorder for performance